*Physics of the Earth and Planetary Interiors*, 62 (1990) 82–96
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

# Paleomagnetic studies on sedimentary Jurassic rocks from southern Bulgaria

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(Received and accepted August 11, 1989)

ABSTRACT

Kruczyk, J., Kądziałko-Hofmokl, M., Nozharov, P., Petkov, N. and Nachev, I., 1990. Paleomagnetic studies on sedimentary Jurassic rocks from southern Bulgaria. Phys. Earth Planet. Inter., 62: 82–96.

Paleomagnetic investigations were performed on Jurassic sediments sampled in the mobile area of Bulgaria comprising Srednogorie, Kreiste and Stranja. The main carriers of magnetic properties of rocks studied are pigmentary haematite and goethite of post-sedimentary origin. The characteristic component of natural remanence (CARM), isolated by means of analysis of demagnetization procedure is secondary, and was acquired after the main tectonic event that took place in this area during the Upper Cretaceous. The CARM directions before tectonic correction are close to the results for Upper Cretaceous magmatic rocks from Srednogorie. The mean CARM direction and the corresponding polar position obtained for the whole region studied are compared with reference data for the Eurasian Platform. The difference between the reference and our data implies anticlockwise rotation of southern Bulgaria relative to Eurasia by ~ 10–20° and a northward tilt of the region under study by ~ 10–15° after acquisition of CARM.

## 1. Introduction

This paper describes paleomagnetic studies on sedimentary Jurassic rocks from southern Bulgaria and their tectonic interpretation relative to the evolution of the mobile part of Bulgaria.

Previously performed paleomagnetic studies of the Jurassic Bulgarian sediments concerned Stara Planina (Balkan). The results indicated that this unit, which is a part of the orogenic Balkanides range, did not undergo rotation with respect to the stable Eurasian plate (Kruczyk et al., 1986, 1988). The present work is concerned with southern Bulgaria, the area comprising Srednogorie and the adjacent smaller units of the Stranja mountains in the west and the Kreiste mountains in the east.

In the Srednogorie and Stranja area so far, paleomagnetic studies have been carried out on Upper Cretaceous magmatic rocks (Nozharov and Veljovich, 1983; Nozharov and Petkov, 1984; Nozharov et al., 1984, 1987). The results of these studies indicate the possibility of post-Cretaceous rotations of this area with respect to the Eurasian platform.

The results presented here are a contribution to the knowledge of the tectonic evolution of Bulgaria as part of the Balkan Peninsula.

## 2. The tectonic characteristics of southern Bulgaria

Bulgaria lies in the northern part of the Tethys area, at the eastern edge of the Mediterranean (Fig. 1). The Mesozoic tectonic evolution of the whole of this area is governed by the interaction of the Eurasian and African plates. Bulgaria is formed of the following tectonic units (Fig. 2a); the Moesian Platform to the north, the Balkanides (sensu lato) in the center and the Rhodopian Massif to the south. The Balkanides (sensu lato)

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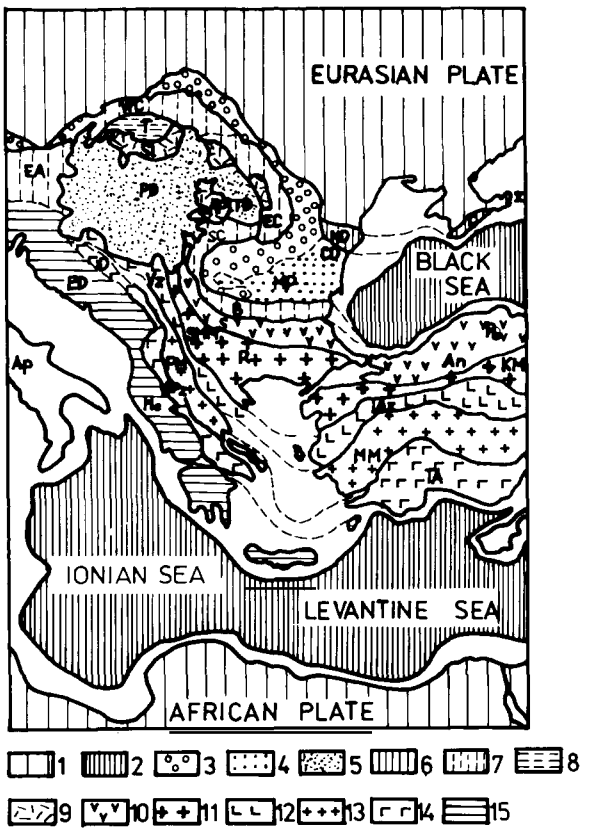



Fig. 1. Tectonic provinces of East Mediterranean area. 1 — External forelands; 2 — present oceans; 3 — external molasse trough; 4 — Moesian Platform (MP); 5 — internal depression: Panonian (PD) and Transylvanian (TD); 6 — North Dobrogea (ND), Central Dobrogea (CD); South Crimea (C); 7 — Eastern Alps (EA), West Carpathians (WC), East Carpathians (EC), South Carpathians (SC) and Fore-Balkan and Balkan (B); 8 — Tatride zone (T); 9 — Slovakia (SI) and Apusini zone (AP); 10 — Srednogorie (S), Pontides (Po), Anatolides (An); 11 — Serbo-Macedonian (SMM), Rhodope (R) and Kirsehir (KM) massifs; 12 — North ophiolite belt: Vardar zone (Vz) and Izmir–Ankara zone (IAz); 13 — Pelagonian (PM), and Mederes (MM) massifs; 14 — South ophiolite belt: Internal Dinarides (ID), Sub-Pelagonian zone (SPz — Pindus zone) and Taurides (TA); 15 — External Dinarides (ED) and Hellenides (He — Apulian zone) (after Hsü et al., 1977).

are divided into the Fore-Balkan, Balkan (Stara Planina) and Srednogorie. Srednogorie, together with the Kraiste and Stranja mountains, forms the region called southern Bulgaria, which is the subject of this work. Southern Bulgaria is separated from the Rhodopian Massif by the deep Maritsa fault, which is a neointrusive zone, and from the Moesian Platform by the deep Balkan–Jambol fault. Figure 2b shows the fracture zones present in Bulgaria as determined from geological data and the isolines of the Moho boundary depth obtained from geophysical data (Dachev and Petkov, 1978).

There are two theories on the tectonic evolution of Bulgaria. According to one (Bončev, 1986), Bulgaria is composed of two rigid units: the Moesian Platform and the Rhodopian Massif (or, more generally, the Pontocaspian and Thracian–Anatolian plates), separated by a collision zone. Between these two rigid plates are the Balkanides with Srednogorie. The Rhodopian Massif is moving northwards, while the Moesian Plate is moving southwards, undergoing at the same time quasi-subduction under the Rhodopian Massif. The other theory (Hsü et al., 1977; Nachev, 1986; Nachev, 1990) attributes most significance to the Vardar subduction zone (Fig. 2a–c). Hsü and Nachev argued that the Balkanide orogen appeared on the active margin of the subduction zone within which the oceanic crust of the Vardar micro-ocean was submerged underneath the tectonic units of Bulgaria. Below we present, in brief, the paleodynamic history of Bulgaria according to this concept.

During the Jurassic in Bulgaria conditions of continental and shallow-sea sedimentation dominated. In this period Africa and Europe moved away from each other, causing a regime of extension to occur between the plates. The Neo-Tethys Ocean and a number of synclines, including the Vardar geosyncline, were formed.

In the Cretaceous the two plates began to approach each other and eventually collided. A compression regime began to dominate, leading to the formation of an Alpine range (sensu lato). In the collision area there appeared a number of microcontinents which, in most general terms, underwent rotations counter-clockwise relative to Europe and other shifts and rotations relative to each other. The main phase of tectonic events in Bulgaria took place in the Middle and Late Cretaceous as a result of the Alpine orogeny (the Austrian stage). In this period Srednogorie remained a eugeosynclinal sedimentary basin (a

back-arc basin according to the plate-tectonic model) with andesite–basalt volcanism, typical of the volcanic activity behind an island arc; here the Rhodopian range (Fig. 2c). The persistence of the compression regime led to the transformation of the marginal basin of Srednogorie into a folded belt. According to Nachev (1990), this occurred before the Maastrichtian.

The Cretaceous tectonic processes caused the distinctly northern vergence of folds throughout the Balkanides and Srednogorie (Ivanov, 1984; Bončev, 1986).

## 3. Sampling procedure

Hand-samples were collected from five sections of the Lower Jurassic (Pliensbachian, Toarcian) and the Middle Jurassic sediments (Bajocian, Bathonian and Callovian) of southern Bulgaria: from Zablano in Kraiste, Gradec and Zmeevo in Srednogorie, and Botevo and Bliznak in the Stranja mountains. From each hand-sample, four to six cylindrical samples were cut in the laboratory for natural remanent magnetization (NRM) measurements and studies of the magnetic properties. The material from Zmeevo and Botevo appeared to be useless for the study and these two localities are not discussed below.

From the Zablano section (ZA) five hand-samples of Pliensbachian quartzite sandstone with limestone matrix were collected from strata tilted at 40–35° to the SSE. After a sedimentation break and an overlying conglomerate layer of about 1-m thickness, Bajocian sandy limestones occurred, tilted at 20–25° to NE, from which three hand-samples were collected.

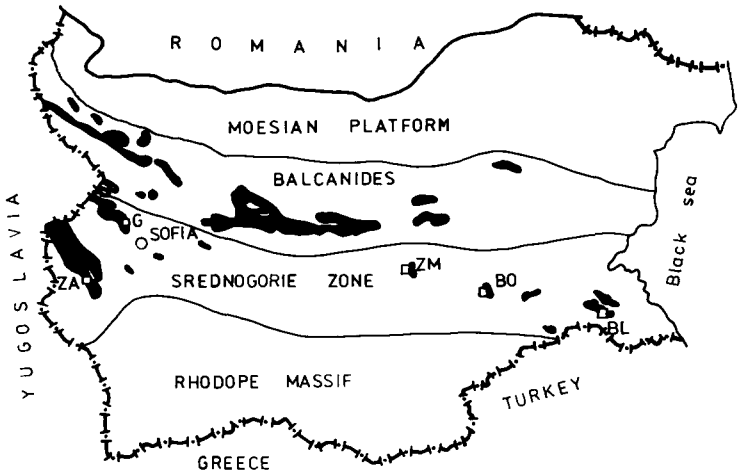
The Gradec section (G) included strata from the Pliensbachian to the Callovian. The Pliensbachian and Pliensbachian–Toarcian strata, built from detrital limestones with oolites, were tilted at 10–15° to the SSE; six samples were collected from them. Bajocian quartzite sandstones (three samples), Bathonian sandy limestones (two samples) and micritic Callovian limestones (three samples) were tilted at 25° to the SW (Bajocian) and 15–20° to the SW (Bathonian and Callovian).

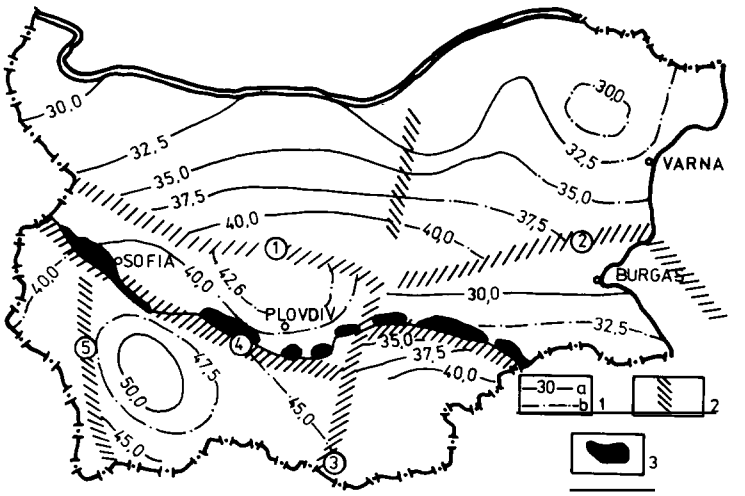
The Bliznak section (BL) covered the Pliensbachian (four samples) Toarcian (three samples), Bathonian (three samples) and Callovian–Kimmeridgian (three samples). The strata were formed from detrital limestone with oolites occuring in inverted layers tilted at 15° to the N in the case of the Pliensbachian, and at 15° to the NNW in the case of the Toarcian. The Bathonian limestones are tilted at 30° to the NNE, and the tilt of the Callovian–Kimmeridgian crystalline limestones is 33° to the N.

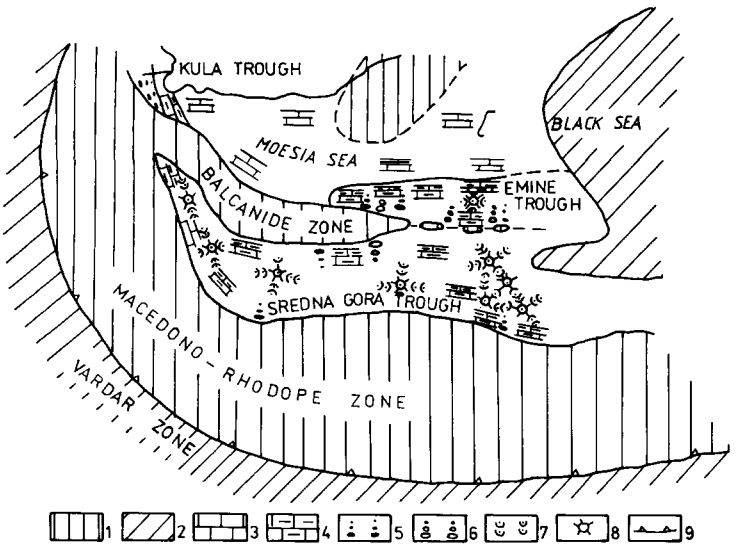
## 4. Experimental methods

The experimental studies were carried out at two paleomagnetic laboratories: at the Geophysical Institute of the Bulgarian Academy of Sciences and at the Geophysical Institute of the Polish Academy of Sciences. The NRM was measured by an IR-4 spinner magnetometer (noise level $5 \times 10^{-6}$ A m$^{-1}$) in Warsaw and Sofia. The magnetic susceptibility $K$ and its anisotropy at room temperature were measured by a KLY-2 bridge (noise level $4 \times 10^{-8}$ SI). In Warsaw, samples were demagnetized by an alternating field (AF) with peak amplitude of 0.08 T, and thermally to 600–700 °C. Thermal and AF cleaning were performed in a practically field-free space, i.e. in a field not exceeding 10 nT. In Sofia the Thellier method was applied.

Fig. 2. (a) Main tectonic units of Bulgaria with outcrops of Jurassic rocks shown as black patches and sampling sites in Srednogorie zone. G — Gradec, ZA — Zablano, ZM — Zmejewo, BO — Botewo, BL — Bliznak. (b) Main deep faults in Bulgaria, based on geological data and isolines of the Moho boundary from seismological data, 1 — M isolines (a, unambiguously identified; b, ambiguous); 2 — deep faults (1, Balkan fault; 2, Jambol fault; 3, Central–Rhodopian fault; 4, Maritsa fault; 5, Strum fault); 3 — Maritsa new intrusive zone (after Datchev and Petkov, 1978). (c) Late Cretaceous (Coniacian) paleogeodynamics of Bulgaria. 1 — Continental crust (emerged lands); 2 — oceanic crust; 3 — shallow-water platform limestones; 4 — pelagic limestones; 5 — siliclastic turbidites (graywackes, siltstones); 6 — Calciclastic turbidites (clastic limestones); 7 — tephraturbidites (tephroidal rocks); 8 — volcanism with lava; 9 — oceanic subduction. (After Nachev, 1990).


R O M A N I A
MOESIAN PLATFORM
BALCANIDES
SOFIA
SREDNOGORIE ZONE
RHODOPE MASSIF
GREECE
TURKEY
Black sea
YUGOSLAVIA


VARNA
BURGAS
SOFIA
PLOVDIV
30,0
32,5
35,0
37,5
40,0
42,6
45,0
47,5
50,0


KULA TROUGH
MOESIA SEA
BLACK SEA
EMINE TROUGH
BALCANIDE ZONE
SREDNA GORA TROUGH
MACEDONO - RHODOPE ZONE
VARDAR ZONE

The magnetic minerals were mainly identified by thermal demagnetization of the isothermal remanence ($I_{rs}$) obtained in a field of 1.8 T at room temperature as the samples were heated to 650 °C in air. The experiments were performed in a field-free space (ensured by a five-layer Permalloy screen). The experiment was performed twice for each sample; the $I_{rs}$ vs. $T$ curves of the second heating show changes in magnetic mineralogy caused by heating in air samples in their natural state. In addition, AF and thermal demagnetization of NRM and changes in susceptibility after successive heating steps were used. For chosen samples representative of particular lithofacies, mineralogical analysis was carried out on polished sections in reflected light using an ore microscope with maximum magnification 1000 ×, by J. Siemiątkowski from the Wrocław Geological Institute.

The NRM demagnetization results were analyzed using Zijderveld orthogonal plots, stereographic plots and the numerical method of Kirschvink (1980). This last method makes possible the calculation of different components of NRM from vectors subtracted during demagnetization procedures, between consecutive cleaning steps. As a result, Kirschvink's method gives directions and intensities of those components together with the accuracy of their determination. One of those components, usually obtained from the greatest number of vectors subtracted, was identified as the characteristic component of NRM (CARM) for each particular sample. Most often this component had the highest intensity and the greatest accuracy of determination.

The mean directions were calculated by the standard methods of Fisher statistics.

## 5. Experimental results

### *5.1. Magnetic minerals*

From optical studies, it was found that the investigated samples of limestones and sandstones had features of rocks which had undergone post-sedimentary transformations. In all the samples the main magnetic minerals were cherry-red goethite or hematite pigment, with grains smaller than 1 μm. Presumably, original chamoisite or chlorite oolites and/or siderite concentrations underwent hematitization or turned into hydrous iron oxides. The pigment was concentrated mainly along cracks and fractures, but also in the matrix, giving the cherry-red colour to the rocks. Detrital remnants were also often tinted with submicroscopic cherry-red pigment. In a few cases, single grains of partially marthitized automorphous magnetite were observed, varying in size from 10 to 100 μm.

The optical studies suggest that the hematite and goethite pigments and the accessory magnetite probably arose because of hydrothermal processes that followed the volcanic activity in this region (Ivanov, 1984; Bončev, 1986; Nachev, 1990).

Results of studies of the magnetic properties of this material agree well with the results of optical investigations. Thermomagnetic analysis of the samples from ZA (Fig. 3a) indicated that goethite and hematite occur in Pliensbachian sandstones ($T_b \sim 100$ and 600 °C for curve 1′), and mainly goethite is present in Bajocian limestones ($T_b \sim 100$ °C for curve 1). The curves of $K$ vs. $T$ show a drop in susceptibility at 300–500 °C, probably related to dehydration of goethite (O'Reilly, 1984) (Fig. 3a).

In the samples from G, on the basis of thermomagnetic analysis, mainly hematite is present (curves 1′ and 1″ in Fig. 3b). Only occasionally does fine-grained magnetite occur in Bajocian sandstones (see curve 1 in Fig. 3b). The susceptibility varies with temperature in various ways — in limestones it does not show any great change up to 300 °C in some samples, and up to 500 °C in others, then increases rapidly as a result of the formation of new magnetic minerals. In sandstones with magnetite it decreases to ~ 30% at 400 °C, then increases.

Thermomagnetic analysis of the samples from BL shows some differentiation of the magnetic fraction (Fig. 3c). Only hematite occurs in samples of detrital (Pliensbachian) limestones (curve 1′), whereas in oolitic limestones (Toarcian) these curves suggest the presence of goethite and a

magnetic mineral with blocking temperatures of ~ 250 °C (curve 1). We suppose that this is submicroscopic magnetite. Bathonian and Callovian–Kimmeridgian limestones contain very small amounts of magnetic minerals which cannot be identified.

Table 1 summarizes the data on the lithofacies investigated, results from optical and thermomag-

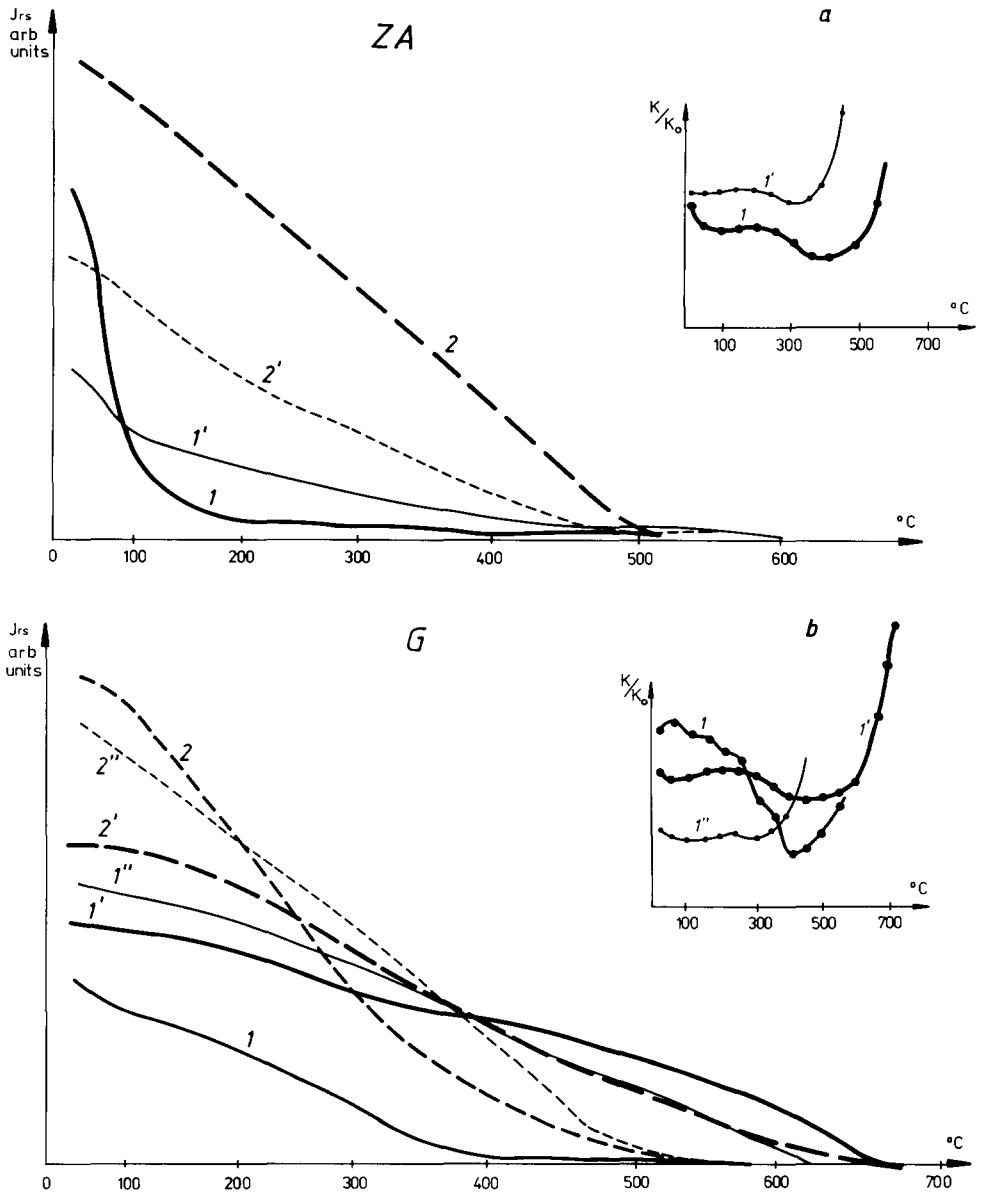



Fig. 3. Thermomagnetic curves measured in the air and changes of susceptibility $K$ measured after each heating step. Thermomagnetic curves of the first heating and curves $K/K_0$ vs. $T$ are denoted by 1, 1′, 1″; thermomagnetic curves of the second heating are denoted by 2, 2′, 2″ respectively. $I_{rs}$ — isothermal remanence acquired in 1.8 T. $K_0$ — susceptibility in the room temperature. (a) ZA: 1 and 1′, limestones, (b) G: 1, sandstone; 1′, detrital limestone with oolites; 1″, limestone, (c) BL: 1, oolitic limestone, 1′, detrital limestone.

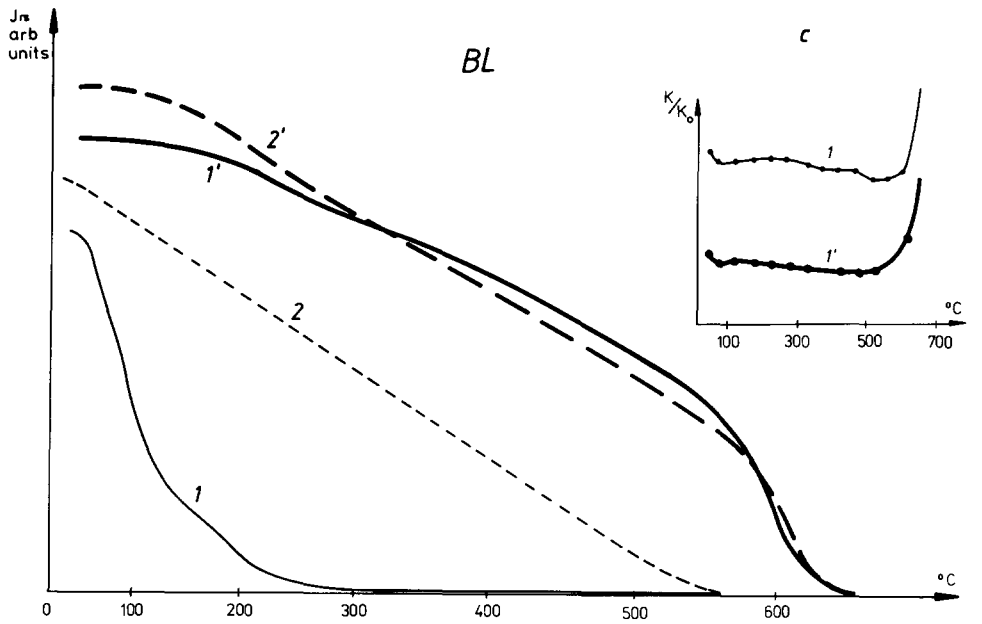



Fig. 3. (continued).

TABLE 1

Lithology and magnetic properties

| Section | Age | Lithology | Magnetic mineralogy | | $K$ ($\times 10^{-6}$ SI) | NRM ($\times 10^{-4}$ A m$^{-1}$) |
|---|---|---|---|---|---|---|
| | | | Optical | Thermomagn. $T_b$ (°C) | | |
| Bliznak (BL) | Callovian–Kimmeridgian | Crinoid limestone | Pigment, small amount | Unidentified | −6−−8 | 0.1–4 |
| | Toarcian | Oolitic limestone | Pigment, hematite or goethite, magnetite | Goethite, magnetite(?) 100–150 °C, 250 °C | 240–370 | 60–100 |
| | Pliensbachian | Detrital limestone | Submicroscopic cherry-red pigment | Hematite 650 °C | 3000–6000 | 250–1000 |
| Gradec (G) | Callovian | Limestone | Cherry-red pigment, hematite or goethite | Hematite 620–670 °C, hematite | 2–8 | 0.8–1.6 |
| | Bathonian | Sandy limestone | | | 15–30 | 0.3–16 |
| | Bajocian | Sandstone | Accessory magnetite | Magnetite 400 °C | 2–30 | 1–2 |
| | Toarcian | Sandstone with oolites | Hematite or goethite | | 300 | 100 |
| | Pliensbachian–Toarcian | Oolites with detrital limestone | Cherry-red pigment | Hematite 620–670 °C | 100–300 | 6–10 |
| | Pliensbachian | | | | 55–145 | 8–12 |
| Zablano (ZA) | Bajocian | Sandy limestone | | Goethite 100–150 °C | 30–55 | 4–15 |
| | Pliensbachian | Quartzite sandstone | Cherry-red pigment, hematite or goethite, hematite grains | Goethite + hematite 100 °C, > 600 °C | 25–45 | 40–80 |

netic studies, and values of the mean susceptibility $K$ and NRM intensity.

### 5.2. *Results of paleomagnetic studies*

Table 2 lists the results of paleomagnetic investigations of the sections studied; mean directions of CARM before (BBC) and after (ABC) tectonic correction, appropriate pole positions, and parameters of the Fisher statistics.

(1) From the profile ZA, seven Pliensbachian and 10 Bajocian samples were demagnetized (mainly thermally). The results obtained are very similar; the NRM intensity decreases to 10–25% of the initial value as the samples are heated to 50 °C (Fig. 4), which indicates that the dominant part of their NRM is related to goethite. The intensity of the remaining, smaller part of the NRM, related to hematite, demagnetizes (usually smoothly) as the temperature increases.

The demagnetization results shown in Zijderveld diagrams (Fig. 4), and the directions of the subtracted components, indicate that in many samples one component is observed. It appears in almost the whole demagnetization temperature range, sometimes including even the initial NRM. This component was identified as the characteristic CARM of this section, carried by goethite and hematite, which must have formed at the same time or over a narrow time interval.

The distribution of the CARM directions will be discussed on the basis of the stereographic projections (Fig. 5a,b), which represent the joint results obtained for the two strata studied. They show the directions of the CARM, before (BBC) and after (ABC) tectonic correction. It can be seen from the distributions presented that tectonic correction causes an improbable separation of the Pliensbachian and the Bajocian directions.

As the composition of the magnetic fraction in the sediments of the horizons studied is similar and the BBC directions are also the same, it can be concluded that carriers of the magnetic remanence — goethite and hematite — were formed after the folding. The close grouping of the directions indicates that the CARM was obtained over a narrow time interval and at the same time in both horizons.

TABLE 2

Paleomagnetic directions and pole positions for southern Bulgaria

| Section | Age of sedimentation | Site | $N/N_1$ | BBC | | | | | | | ABC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $D_m$ | $I_m$ | $\alpha_{95}$ | $K$ | $\Phi_N$ | $\Lambda_E$ | $\varphi_m$ | $D_m$ | $I_m$ | $\alpha_{95}$ | $K$ | $\Phi_N$ | $\Lambda_E$ |
| Bliznak (BL) | Callovian– | | | | | | | | | | | | | | | |
| φ = 42.2 °N, λ-27.5 °E | Kimmeridgian | 1 | 7/12 | 347 | 54 | 9.0 | 46.3 | 78 | 265 | 35 | 351 | 22 | 9.0 | 46.0 | 58 | 224 |
| Stranja Mts | Toarcian | | | | | | | | | | | | | | | |
| | Pliensbachian | 4 | 16/19 | 348 | 65 | 4.5 | 66.8 | 80 | 332 | 47 | 341 | −83 | 4.8 | 59.0 | 29 | 32 |
| Gradec (G) | Callovian | | | | | | | | | | | | | | | |
| φ = 42.7 °N, λ = 23.2 °E | Bathonian | | | | | | | | | | | | | | | |
| | Bajocian | 3 | 13/14 | 345 | 69 | 7.7 | 299 | 76 | 340 | 53 | 296 | 67 | 7.7 | 30.1 | 46 | 326 |
| Western Srednogorie | Toarcian | | | | | | | | | | | | | | | |
| | Pliensbachian | 2 | 13/20 | 325 | 65 | 8.3 | 26.1 | 65 | 314 | 47 | 293 | 75 | 7.4 | 35.3 | 47 | 343 |
| Zablano (ZA) | Bajocian | | | | | | | | | | | | | | | |
| φ = 42.5 °N, λ = 23 °E | | 2 | 7/7 | 345 | 63 | 5.5 | 119.1 | 79 | 307 | 44.5 | 13 | 50 | 5.6 | 118.4 | 74 | 157 |
| Kreiste | Pliensbachian | 2 | 10/10 | 344 | 64 | 8.2 | 35.7 | 78 | 314 | 46 | 158 | 79 | 8.2 | 36.0 | 22 | 32 |
| Mean southern Bulgaria | Jurassic | | 6 | 343 | 64 | 5.2 | 167.3 | 77 | 313 | 45 | | | | | | |

$N_1$ — Number of samples demagnetized; $N$ — number of samples used for calculations; BBC — before bedding correction; ABC — after bedding correction. Mean direction and pole position calculated for φ = 42.5 °N, λ = 25 °E.

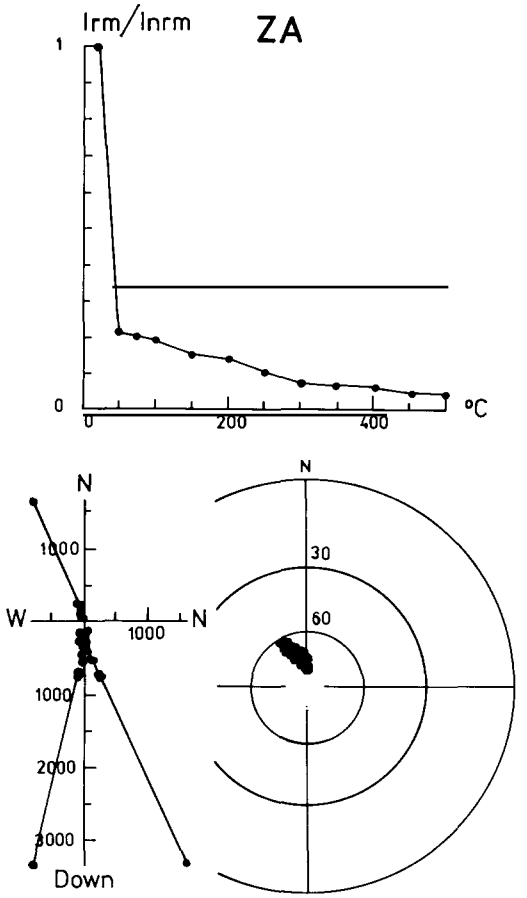



Fig. 4. Results of demagnetization of typical ZA limestone (Bajocian), $I_{rm}$ — intensity of remancence measured after each heating step, $I_{nrm}$ — intensity of NRM before heating. Intensity of components of remanence in A $m^{-1} \times 10^{-6}$.

(2) From the section G, using temperature (for the most part) and AF, 20 samples from the Upper Jurassic and 14 samples from the Middle Jurassic were demagnetized. AF demagnetization was not effective because of the presence of hematite pigment — after demagnetization by a field of 0.09 T, ~ 60% of the NRM still remained. The samples demagnetized thermally were heated to 600–700 ° C. However, it was possible to analyze the demagnetization curves only up to 400–500 ° C, because at higher temperatures mineralogical changes are suggested by susceptibility changes (see Fig. 6).

The Zijderveld diagrams (Fig. 6a,b) show that the NRM of the samples investigated consists of a few (two or three) components. The CARM was identified as the component calculated from the greatest number of subtracted vectors, and usually having the highest intensity. The blocking temperatures of the carriers of this component did not as a rule exceed 300 ° C.

What remains to be considered is the problem of whether the CARM was acquired before or after folding. The tilts in Gradec are similar in the studied horizons of the Lower and Middle Jurassic, and so tectonic correction does not decisively change the mean directions and the statistical parameters (see Fig. 7 and Table 2). The only criterion which determines the choice of the CARM directions before the tectonic correction for further considerations is the origin of the carriers of this remanence. Previously it was concluded that they are secondary, and therefore we assume that the CARM is a secondary remanence, arising after the folding, as in ZA. Thus, we treat the mean direction before tectonic correction as the CARM.

(3) From the section BL, 19 samples from the Lower Jurassic (Pliensbachian and Toarcian) and 16 samples from the Middle Jurassic (Bathonian and Callovian–Kimmeridgian) were demagnetized, mostly thermally. In the 16 samples from the Lower Jurassic, either only one or two NRM components could be identified (Fig. 8). The characteristic component is that whose mean direction, shown in Table 2, is either the only component, or the component obtained from the greater number of vectors subtracted. The CARM directions have low scatter ($\alpha = 4.5°$), which indicates that all the remanence carriers described in the previous section were formed over a narrow time interval. The sediments studied are from inverted strata and the tectonic correction causes a change in the CARM polarity. Moreover, the mean direction gains an improbably high inclination of 83° after tectonic correction. The results discussed are shown in Fig. 9 and Table 2. On the basis of mineralogical analysis and from analysis of the directions before and after tectonic correction, it was concluded that the remanence of the rocks from BL was acquired after folding.

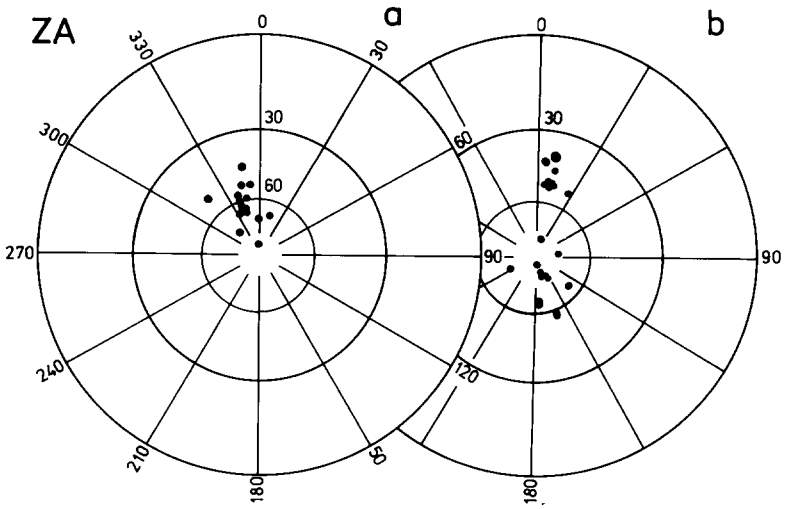



Fig. 5. Stereographic projection of CARM directions for ZA. (a) Before tectonic correction, (b) after tectonic correction.

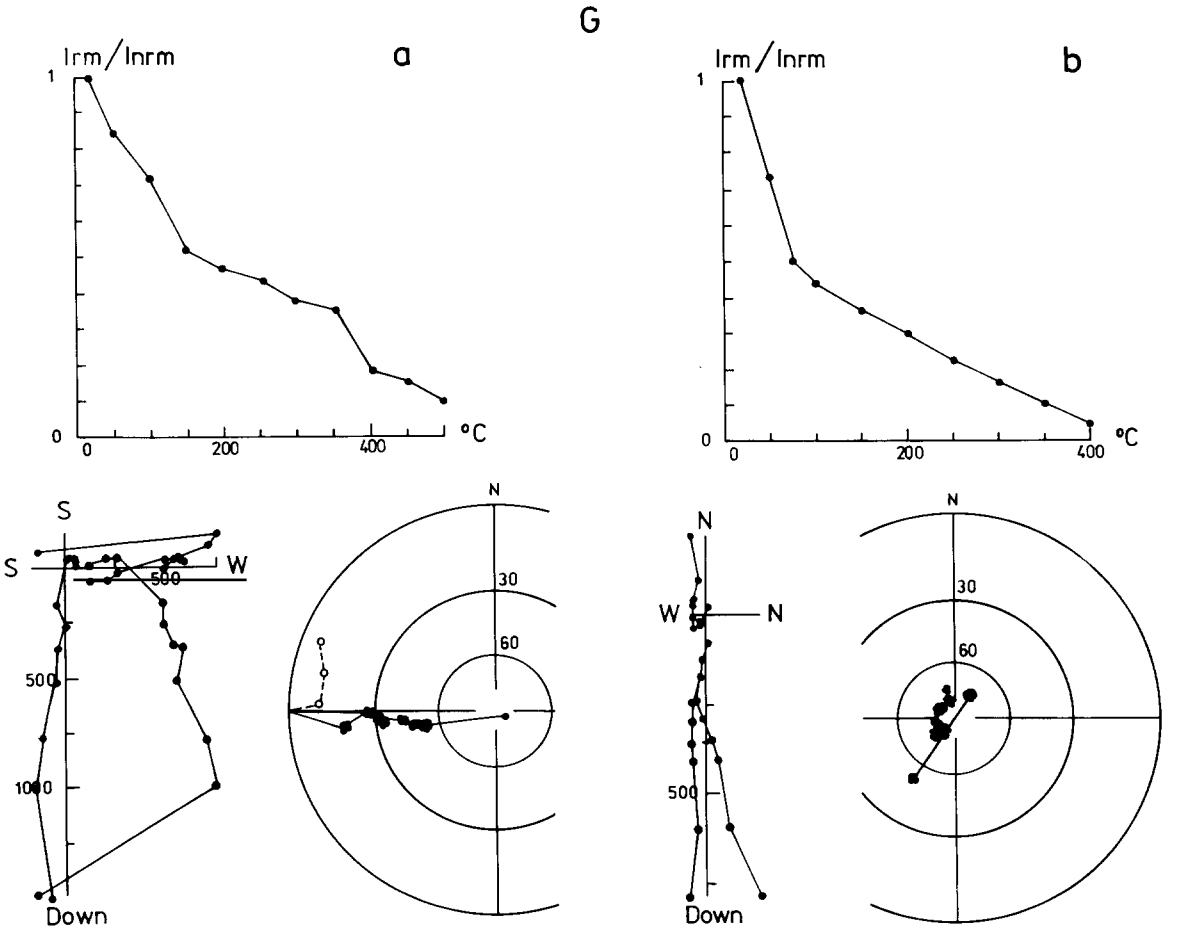



Fig. 6. Results of demagnetization of typical G samples. (a) Detrital limestone with oolites (Pliensbachian), (b) sandstone (Bajocian). For legend see Fig. 4.

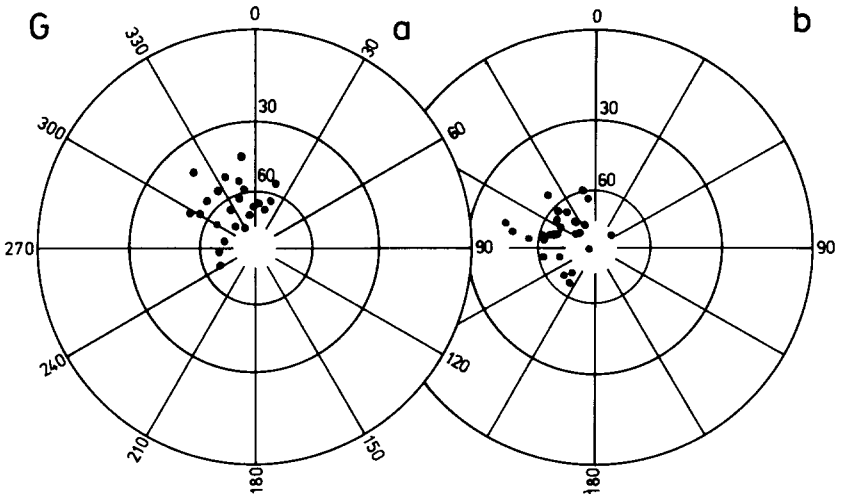



Fig. 7. Stereographic projection of CARM directions for G. (a) Before tectonic correction, (b) after tectonic correction.

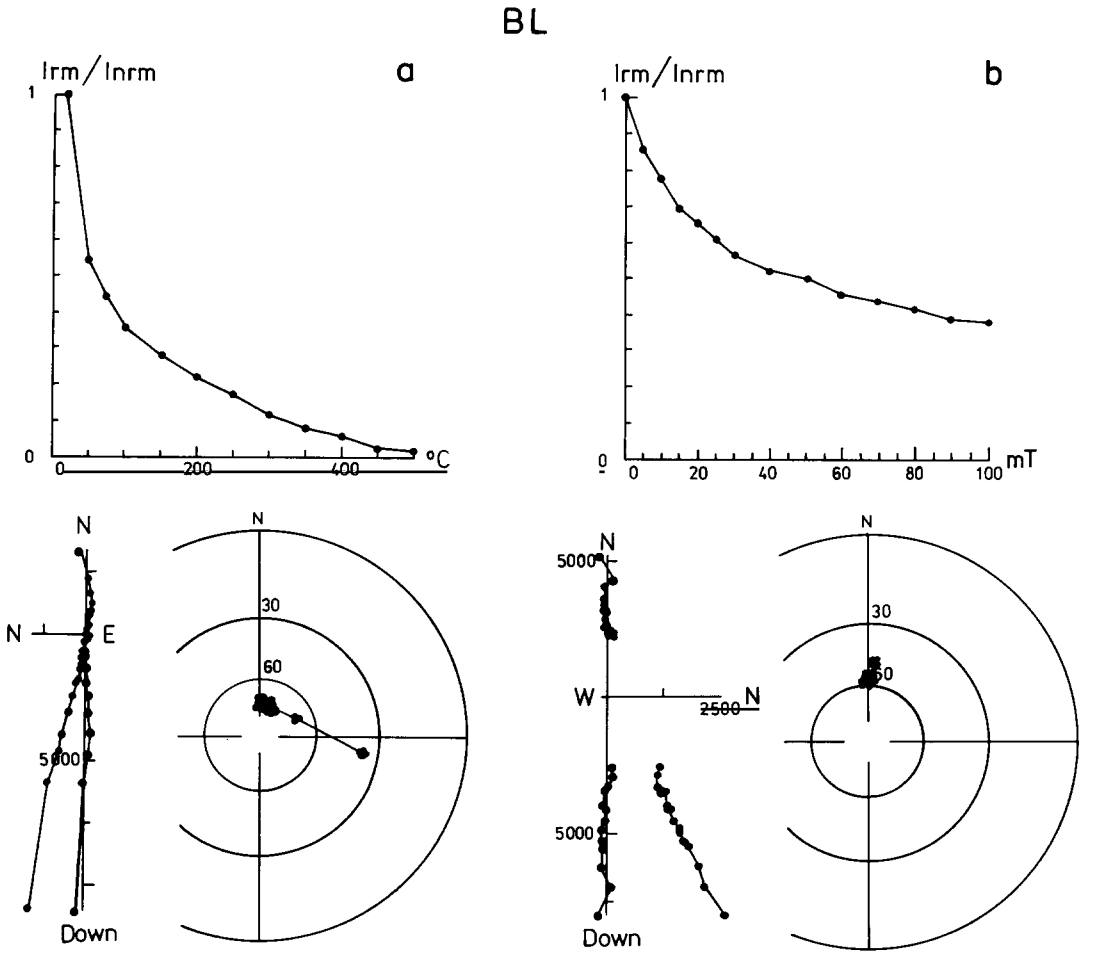



Fig. 8. Results of demagnetization of typical BL samples. (a) detrital limestone with oolites (Pliensbachian), (b) limestone (Toarcian). For legend see Fig. 4.

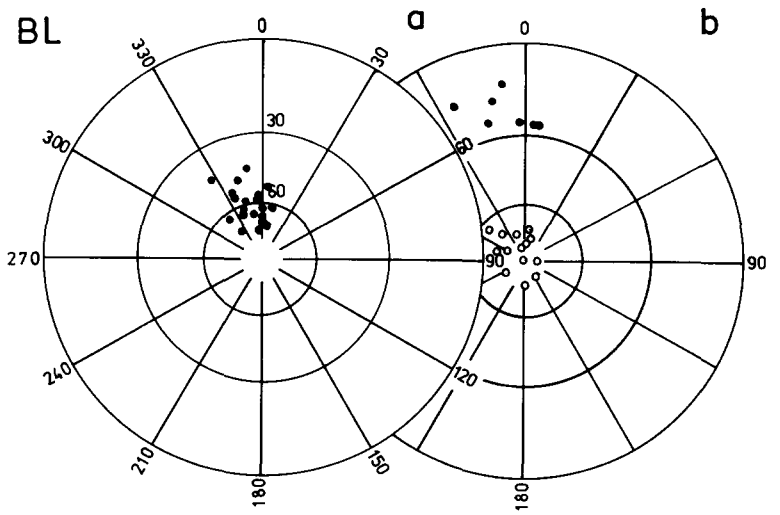



Fig. 9. Stereographic projections for BL. (a) Before tectonic correction, (b) after tectonic correction.

Only four samples were demagnetized from the Bathonian limestones. The directions of their NRM were greatly scattered, and the intensity was very low (see Table 1), so they were not used in further interpretation. The results obtained for seven out of 12 demagnetized Callovian–Kimmeridgian specimens are reasonably well grouped, despite their low NRM intensity and negative $K$, and these results are included in Table 2.

### *5.3. Anisotropy of the magnetic susceptibility*

The anisotropy of the magnetic susceptibility was also measured at room temperature. The results were analyzed by means of the Jelinek Anizo 11 program (Jelinek, 1977), which gave for each specimen studied a set of parameters describing the shape of the susceptibility ellipsoid and the directions of its main axes. It was found that a distinct concentration of the directions of these axes occurs only in the Lower Jurassic limestones from the BL. The axis $K_{min}$ is here directed perpendicular to the sedimentation plane, and the axes $K_{max}$ and $K_{int}$ are directed respectively north–south and east–west. This image, obtained after tectonic correction, indicates that the secondary minerals — the carriers of the magnetic properties — retained the primary, sedimentary magnetic rock structure. In the other cases, no distinct ordering was observed. The shape of the

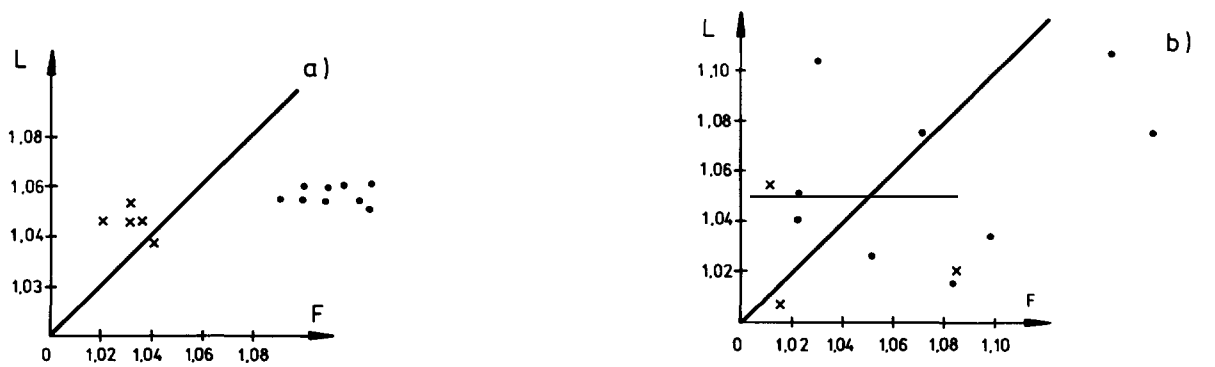



Fig. 10. L–F diagrams. (a) BL limestones: dots — Pliensbachian, crosses — Toarcian, (b) G: dots — Bajocian sandstones, crosses — Bathonian sandy limestones.

susceptibility ellipsoid is defined by the factors $L$ (lineation) and $F$ (foliation); $L$–$F$ diagrams for the Pliensbachian limestones from BL show the distinct domination of foliation (a flattened ellipsoid). In the Toarcian limestones, an elongated $L$–$F$ ellipsoid occurs (Fig. 10a).

In the limestones from G and ZA, the values of $L$ and $F$ are small (not exceeding 1.02), whereas in the samples of sandstones and sandy limestones from G (Bajocian and Bathonian), the values of the two parameters are scattered (Fig. 10b).

## 6. Discussion of the results

The experimental results presented above indicate that the NRM in the Jurassic rocks studied here is secondary and that the direction of the geomagnetic field in which this NRM was acquired is given by the CARM before tectonic correction. For the five groups studied, the mean CARM directions before correction have small scatter, whereas the mean directions for these groups after correction are dispersed so much that it is useless to calculate a mean direction from them (see Table 2). Therefore we believe that the CARM in all three sections under study was acquired during the same time interval after the investigated sediments had been folded. According to Nachev (in press), the filling of the Srednogorie basin, compression and volcanic activity in this region took place before the Maastrichtian. Thus, we think that the secondary magnetic minerals, formed as a result of post-volcanic hydrothermal activity, acquired the magnetic remanence only at the end of the Upper Cretaceous.

To determine more precisely the age of the CARM, the results obtained were compared with those of the paleomagnetic studies carried out by Nozharov and Veljovich (1983), Nozharov and Petkov (1984) and Nozharov et al. (1984, 1987), on the Upper Cretaceous magmatic rocks in southern Bulgaria (Table 3). It can be seen that the mean direction obtained here is in agreement with the cited Upper Cretaceous directions of the geomagnetic field in southern Bulgaria. It is thus suggested that the Jurassic sediments under study acquired the post-folding magnetic remanence

TABLE 3

Compilation of Upper Cretaceous paleomagnetic results for southern Bulgaria compared with the reference Eurasian data

| No. | Region | Age of rocks | Age of NRM | $D_m$ | $I_m$ | $\Delta_{D_m}$ | $\Delta_{I_m}$ | ΦN | ΛE | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | South Bulgaria before bedding correction | Lower–Middle Jurassic | Upper Cretaceous | 343 | 64 | −21 | +14 | 77 | 313 | N |
| 2 | Srednogorie, magmatics calculated from Nozharov et al. (1984) | Upper Cretaceous (Santonian) | Upper Cretaceous (Senonian) | 350 | 60 | −14 | +10 | 82 | 286 | N |
| 3 | Gutzal Plutonium $\varphi = 42.5°$ N, $\lambda = 23.2°$ E (Nozharov et al., 1987) | Campanian | Campanian | 356 | 58 | −8 | +8 | 86 | 243 | N |
| 4 | Eurasia (from reference) curve; Westphal et al., 1986) | Upper Cretaceous | Upper Cretaceous | 4 | 50 | – | – | 78 | 188 | M |

1, 2 and 4 — Calculated for $\varphi = 42.5°$ N, $\lambda = 25°$ N.
2 — No statistical parameters because the mean direction was calculated from only two values.
3 — No statistical parameters in the source paper.
4 — $K = 110$, $\alpha_{95} = 12$.

close to the time of formation of the Upper Cretaceous magmatic rocks dated to the Santonian–Campanian (Nozharov and Veljovich, 1983; Nozharov et al., 1987).

It should be pointed out that the NRM of the Jurassic sediments studied and most of the Upper Cretaceous magmatic rocks (see Nozharov and Veljovich, 1983; Nozharov et al., 1987) had a normal direction, which shows that they were magnetized in a geomagnetic field of normal polarity (N). In the Upper Cretaceous a normal direction of the geomagnetic field dominated. Figure 11 shows that the field N covered periods from the Cenomaninan to the Santonian (part of the so-called Quiet Cretaceous Period) and the upper part of the Campanian. In the Lower and Upper Maastrichtian the Earth's field also had normal polarity.

Considering the above arguments, we believe that the Jurassic sediments studied acquired their NRM in the Upper Campanian or Maastrichtian, during a period of normal polarity, after regional folding.

Table 3 shows the position of the paleomagnetic pole in the Upper Cretaceous for Eurasia (the reference polar wander curve is according to Westphal et al. (1986)) and the expected direction of the geomagnetic field in the Upper Cretaceous in southern Bulgaria as calculated from this pole. The directions obtained here for sedimentary rocks and those found by Nozharov and co-workers for magmatic rocks differ from the reference 'Eurasian' direction for the Upper Cretaceous by ~ 10–20° in declination and have inclination higher by ~ 10–15°. The difference in declination indicates a counter-clockwise rotation of the fragments of southern Bulgaria by 10–20° relative to Eurasia. From the structural and tectonic viewpoint (Nachev, in press), this rotation is very probable, and may have been related to the system of faults caused by the Mesozoic extensive–compressive regime. The higher values of the inclination of remagnetized Jurassic sediments can be explained by assuming that they underwent a process which increased the tilt of the layers as they acquired magnetic remanence. It can be suggested that the same mechanism which tilted the whole of southern Bulgaria in post-Cretaceous time caused

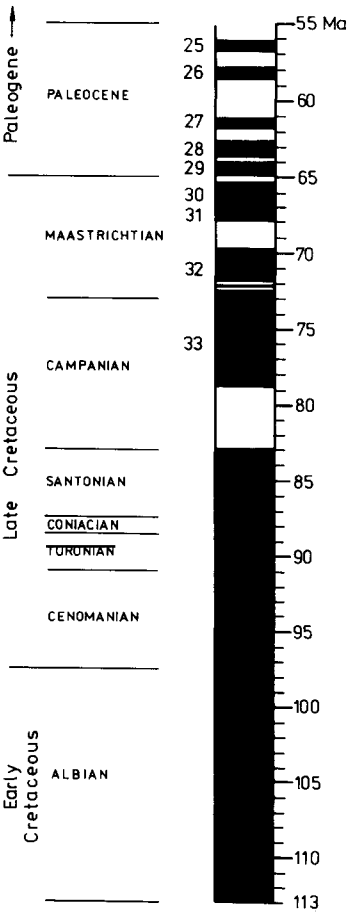



Fig. 11. The geomagnetic polarity time scale from Albian to Paleocene as based on a combination of land and marine studies. Numbers on the left side of the column denote numbers of oceanic anomalies: white intervals–R polarity; black intervals–N polarity (after Piper, 1987).

the increase of inclination of magmatic rocks studied by Nozharov and co-workers. Perhaps the mechanism leading to the uplift of the south Bulgarian block should be connected with the deep sublateral Maritsa fault (Fig. 2b and 2c), and the post-Upper Cretaceous compression regime in the region including the Vardar subduction zone.

## 7. Conclusions

(1) The NRM of the investigated Jurassic sediments in southern Bulgaria is carried by secondary

magnetic minerals. The CARM is a post-folding remanence with normal polarity, acquired at the end of the Upper Cretaceous, probably in the Campanian or one of the Maastrichtian periods of normal polarity of the geomagnetic field.

(2) Southern Bulgaria rotated by $\sim 10-20°$ counter-clockwise relative to the Eurasian Platform. This rotation occurred after the Upper Cretaceous.

(3) The investigated sediments have been tilted by $\sim 15°$ after acquiring NRM, probably as a result of the tectonic activity in this region. We suppose that the same mechanism is responsible for the tilting of the whole of southern Bulgaria.

## Acknowledgement

This work was supported by the Polish Academy of Sciences through grant CPBP 03.02.

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